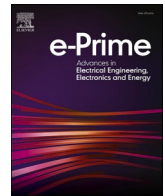




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Analysis of multiple faults in induction motor using machine learning techniques

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ABSTRACT

Studying simultaneous faults in three-phase induction motors guarantees reliability, reduces unplanned downtime, and minimizes maintenance expenses in industrial settings. Induction motors have concurrent faults, including stator winding, rotor faults, unbalanced voltages, load fluctuations, and overvoltages. These make fault diagnosis difficult and can result in disastrous failures if not detected. Traditional diagnostic methods are expert judgment-based and pre-threshold-based and, therefore, less efficient when dealing with vast industrial processes. Based on key operating parameters like voltage, current, and speed, this article describes how machine learning (ML) algorithms like Random Forest (RF), K-Nearest Neighbors (KNN), Gradient Boosting Machine (GBM), Support Vector Machines (SVM), and Extreme Gradient Boosting with Feature Interaction (XGBoost + FIS) are used to detect different motor faults. Due to their limits, machine learning algorithms outperform traditional methods in real-time fault diagnosis, predictive maintenance, and multi-fault categorization. Through ensemble learning and feature selection, the models cope well with big data sets with enhanced fault classification accuracy and robustness against noise. In addition, ML fault analysis minimizes reliance on human experience and presents a computerized, scalable industrial motor condition monitoring method. Experimental outcomes show that excellent classification accuracy is achieved in ML models; hence, active maintenance and effective motor operation are feasible. The outcomes point to the potential of AI-based predictive maintenance to enhance safety, energy efficiency, and process continuity in industrial applications. Traditional fault detection methods rely on pre-established thresholds or expert-provided rules and may not perform effectively for concurrent multiple faults. In order to surpass these limitations, a new approach by using state-of-the-art machine learning algorithms such as Extreme Gradient Boosting (XGBoost) combined with Fuzzy Inference Systems (FIS) presents a new perspective towards improved accuracy and comprehensibility in fault detection. This method takes advantage of the ability of XGBoost to learn intricate relationships and utilizes FIS's rule-based reasoning for explainability. The incorporation of FIS in XGBoost enhances the precision of fault classification, handles uncertainty, and encourages interpretability.

1. Introduction

Owing to their reliability, price competitiveness, and energy efficiency, three-phase induction motors are employed heavily across industry. Several manufacturing, energy, transportation, and mining sectors form the backbone of industrial power machinery and automation. Reliable as they may be, these motors are infested with numerous faults, resulting in production downtime, high maintenance expense, and potential safety risks [1]. Unintermittent and error-free operation is

a critical aspect of industrial efficiency for three-phase induction motors. While good in some applications, classical methods for fault detection are unable to handle complex multi-fault scenarios. As industries move towards automation and data-driven decision-making, machine learning (ML) methods have proved useful for enhancing electric motor fault detection and predictive maintenance.

Fault diagnosis and detection of three-phase induction motors are important for industry maintenance planning. Induction motor fault can result in expensive outcomes, including unplanned downtime, loss of

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efficiency, and, at worst, catastrophic failure with safety risks. The most common faults in induction motors include stator winding faults, rotor faults, unbalanced voltages, oscillations of loads, and overvoltages [2]. These deficiencies can lead to cascading failures and cause major industrial processes to be disturbed if not detected early. Motor faults have been traditionally diagnosed through methods such as vibration analysis, infrared thermography, model-based techniques, and motor current signature analysis (MCSA) [3]. They are labor-intensive, error-prone, and rely on experienced personnel for correct interpretation and diagnosis since they heavily depend on pre-specified threshold values, manual feature extraction, and human expert knowledge.

Additionally, they do not perform well in handling intricate fault patterns, changing operating conditions, and detecting defects in the initial stages. With the progression of industrial automation over time, such conventional methods fail to cope with high-volume motor health monitoring effectively. Intelligent data-based techniques that allow for more accurate and reliable fault detection are increasingly sought after. They are generally less precise and reactive under traditional conditions in industrial setups when motors are subjected to fluctuating loads and environmental scenarios. The need for efficient fault classification and predictive maintenance under such situations prompted researchers and engineers to look towards sophisticated machine learning-based methods [4].

Diagnosis is complex for faults in three-phase induction motors since fault signatures overlap, various operating conditions, and real-time monitoring complexity under complex industrial conditions [5]. Most motor faults have the same symptoms; hence, it's hard to distinguish them based on traditional methods. Most conventional fault detection techniques also require reading with much expert knowledge, which is challenging to scale up to industrial levels. Fault detection delay caused by manual inspection and threshold-based diagnosis methods typically fail later, creating high-cost downtime. In addition, it is not simple to track induction motors in real time because sensor-generated data is generated, which must be processed using sophisticated methods to process efficiently [6]. Such problems can be avoided using advanced machine learning algorithms to process big data, differentiate between faults, and perform real-time fault classification for proactive maintenance.

The conventional fault diagnosis methods of induction motors also have a number of disadvantages, such as not being able to diagnose multiple faults at a time, being based on human experience to understand the results, and being inefficient in processing bulk data. Furthermore, conventional methods are unable to provide real-time operation because their pre-specified thresholds are not flexible to vary with various motor conditions [7]. These methods are also expensive as they use high-cost equipment sensors and frequent manual monitoring. These considerations suggest the necessity for more advanced means to utilize artificial intelligence to enhance fault classifying effectiveness and identify in real-time.

Machine learning has revolutionized fault detection and induction motor diagnosis with data-driven methods that learn automatically from plant data. In contrast to conventional methods, ML models can generalize to novel data and operate very well for real-time fault diagnosis [8]. ML models utilize current, voltage, speed, and temperature sensor measurements to detect anomalies and types of faults efficiently. Applying ML models can help companies reduce downtime, lower maintenance costs, and increase the operating capacity of motors. These features involve automatic feature selection, high-accuracy classification, adjustment of operating conditions, and industrial application suitability.

Because of its accuracy and stability, Random Forest (RF) is a powerful machine-learning method extensively used in induction motor fault diagnosis classification. RF is an ensemble learning method, where multiple decision trees are constructed during training and their estimates are averaged for better classification performance, reduced variance, and prevention of overfitting. The method increases the reliability

of fault detection by utilizing diverse decision paths and is, therefore, ideal for handling noisy data and complex fault patterns. RF's feature importance ranking ability is also helpful in pinpointing important parameters affecting motor faults, facilitating more interpretable and data-driven fault diagnosis in industrial contexts. A non-parametric technique called K-Nearest Neighbors (KNN) categorizes errors according to the separation between points in the feature space. Support Vector Machines (SVM), a supervised learning approach, generate the optimal hyperplane for fault class differentiation. By improving the present classifier and using difficult-to-classify examples to create new classifiers that correct past errors, the Gradient Boosting Machine (GBM) boosting algorithm generates successive approximations of the best classifier. Extreme Gradient Boosting (XGBoost) with Feature Interaction (FIS) is a combination model that unifies the high performance of XGBoost with fuzzy inference systems' explainability (FIS). These machine learning techniques have been found to work effectively in fault detection and classification through statistical learning, pattern recognition, and feature selection techniques.

One of the more recent developments in fault diagnosis has been merging FIS with machine learning. Although ML models are highly accurate at classification, they are black-box models, which restricts interpretability. Conversely, FIS uses rule-based reasoning to provide human-interpretable information about the fault diagnosis process. The XGBoost + FIS hybrid model combines the predictive power of ML and the interpretability of fuzzy logic to provide an improved industrial fault classification solution. Incorporating interpretability in fault classification allows industries to build a richer fault state understanding, which translates into enhanced decision-making and maintenance practices.

Industrial motor condition monitoring using machine learning is fast gaining traction due to its ability to reduce unplanned downtime and enhance operating effectiveness. Fault detection systems using ML are used in manufacturing facilities to monitor motor conditions and anticipate failures before failure [9]. ML-based models analyze sensor data from motors and turbines in power stations to improve efficiency and reduce maintenance expenses. The transportation sector applies machine learning to detect electrical motor faults in trains, trucks, and conveyor belts. Steel and mining applications involving heavy loads utilize ML-based predictive maintenance that prevents expensive breakdowns. ML-based monitoring also plays a gigantic role in fault detection in the electricity grid and optimizing energy delivery.

The main aim of this project is to design an improved three-phase induction motor fault detection system based on machine learning models. This involves critically evaluating classification models, their failures, and frequent revisions to enhance them to be more effective, accurate, and interpretable. The research is aimed at fault diagnosis for the motor faults of stator and rotor, bearing faults, and voltage unbalance using sensor-based machine learning models. Leveraging advanced algorithms such as ensemble learning and deep learning, the system aims to enhance predictive maintenance, reduce downtime, and maximize the overall industrial motor operation reliability. This entails extensively examining classification models and their limitations and iterative optimization to enhance them to be more accurate, effective, and interpretable. The research also seeks to determine the drawbacks of conventional models and alleviate them using ensemble learning and boosting methods to offer greater fault classification accuracy. Besides, to offer an interpretation of the models, the research combines Fuzzy Inference Systems (FIS) and XGBoost to create a hybrid system with high predictability ability along with rule-based reasoning such that fault diagnosis will be more interpretable and reliable for industrial predictive maintenance.

Using advanced machine-learning techniques for induction motor defect detection is necessary for increasingly complex industrial operations. While better than other approaches in the laboratory, current algorithms for multi-fault classification processes do not perform as well under real-world circumstances in industrial settings. Machine learning offers a live, scalable, and dependable platform for motor wellness

monitoring; thus, it makes predictive maintenance accessible to improve working reliability. This paper thoroughly reviews fault diagnosis techniques using ML models, including Random Forest, KNN, SVM, GBM, and XGBoost + FIS, proving the feasibility of AI-based solutions for industrial applications. The outcome of this work can be used to create more fault-tolerant and explainable fault detection systems, leading to better motor reliability and efficiency in different industries.

By presenting an advanced machine learning-based fault classification model for three-phase induction motors, this study significantly advances the field of industrial motor fault detection. Conventional fault detection techniques utilize pre-defined thresholds and human experience and cannot deal with multi-fault cases and large-scale industrial applications. The paper illustrates data-driven methods through machine learning models, including RF, KNN, SVM, GBM, and a heterogeneous Extreme GB with Feature Interaction (XGBoost + FIS) model. The models improve accuracy, reliability, and scalability for motor fault diagnosis by identifying the most significant operating parameter's voltage, current, and speed. One of the essential contributions of this research is the development of the XGBoost + FIS hybrid model, where the best prediction power of Extreme Gradient Boosting (XGBoost) is combined with the interpretability of FIS. This addresses the black-box problem of standard machine-learning models through the ability to produce explainable fault classification, thereby allowing industries to make well-informed maintenance choices. The research verifies that ensemble-based models, especially XGBoost+FIS, surpass standard classifiers with the highest possible accuracy of 95 % in fault classification.

Moreover, this work suggests an excellent feature selection and preprocessing pipeline to ensure optimal model performance with negligible computational complexity. For efficient learning, the set of 1600 labeled samples under various fault conditions is managed with normalization, feature engineering, and stratified splitting. The paper also determines the limitations of conventional classifiers, including SVM and KNN, in handling overlapping fault classes and complex industrial conditions, citing the benefit of ensemble-based learning techniques. By applying machine learning techniques to predictive maintenance, this paper contributes to reducing unplanned downtime, minimizing maintenance expenditure, and increasing the reliability of industrial motors during operation. The results have extensive applications in the manufacturing, power generation, and transportation industries, where real-time fault detection is vital. The research also opens the door for further research into deep learning-based fault diagnosis, IoT-based real-time monitoring, and energy-saving AI models for industrial automation.

The organization of this paper is as follows: [Section 1](#) presents the significance of fault detection in three-phase induction motors, the limitations of conventional diagnostic methods, challenges in real-time monitoring, and the advantages of machine learning-based approaches to improve accuracy, reliability, and predictive maintenance. [Section 2](#) is a literature review of current fault detection methods, contrasting conventional and machine learning approaches and determining research gaps. The experimental setup, dataset, preprocessing, feature extraction, and machine learning classification model, which includes Extreme Gradient Boosting with Feature Interaction (XGBoost + FIS), are all included in [Section 3](#)'s methodology section. The findings of a comparison study of model performance on accuracy, precision, recall, and F1-score are presented in [Section 4](#), which shows that XGBoost + FIS performs better in fault classification. [Section 5](#) concludes the research, highlighting significant findings, stressing the use of machine learning in predictive maintenance, and suggesting future evolution by introducing deep learning and IoT-enabled real-time monitoring. [Section 6](#) provides references cited in this study. Such a systematic approach ensures a thorough investigation of induction motor fault-diagnosis machine learning techniques and how they can be applied to industry.

2. Related work

Machine learning-based techniques for diagnosing electric vehicle (EV) induction motor problems have been presented by certain researchers [10,11]. For example, ANSYS RMxprt was used to simulate a 5 HP, 415 V, 50 Hz sample induction motor. Different machine learning classifiers were used for defect detection, such as SVM, k-NN, MLP, Random Forest, Decision Tree, Gradient Boosting, XGBoost, and Deep Learning. Out of these, Random Forest, Decision Tree, Gradient Boosting, and XGBoost models attained a perfect precision rate of 100 %. However, the accuracy of the SVM and MLP models was low because of misclassification. Although real-time implementation concerns and computational efficiency were not thoroughly investigated, the method proved successful for fault detection in areas such as open-phase circuits (OPC), high resistance connection (HRC), and short circuits (SC).

Comparative studies have also been conducted on fault diagnosis in electric machines based on machine learning methods [11,12]. In [11], fault diagnosis based on deep learning was investigated, and it was found that CNN and RNN performed better than traditional approaches regarding precision and robustness. Based on real-world industrial electric machine datasets, the faults were targeted in the stator windings, rotors, and bearings. Various machine learning models were compared using performance criteria such as F1-score, recall, accuracy, and precision. Using Layer-wise Relevance Propagation (LRP) to make deep learning models more explainable, model interpretability was also investigated. Although the models were more precise, deep learning models consumed lots of computational power and data, making real-time processing difficult. A MATLAB system was suggested for detecting faults like open circuits, short circuits, overload, and faulty rotor bars in induction motors [12]. The Fast Fourier Transform (FFT) and other signal processing techniques were used to process the frequency components, and Decision Trees, SVM, Logistic Regression, and Naïve Bayes were used for classification. With 92 percent accuracy, decision trees were the most accurate. The study highlighted the necessity of combining signal processing and machine learning for industrial problem identification, but it ignored real-time deployment concerns and computation costs.

Researchers have also investigated more advanced classification methods, like empirical mode decomposition and Discrete Wavelet Transform (DWT) for feature extraction [13]. In [13], classifiers like Support Vector Machine (SVM), Decision Tree, Random Forest, and XGBoost were used to detect faults in rotor bars, bearings, and interturn faults. The experimental setup was motor current signature analysis and time-frequency domain signal feature extraction. It was found that ensemble models like Random Forest and XGBoost performed well under multi-fault scenarios, but Decision Trees proved to be most effective under a single-fault scenario. The research concluded that the choice of classifier is based on computational power and real-time requirements, which are areas of further study. Several studies have classified induction motor faults using machine learning models using experimental data [14,15]. Seventeen classifiers from MATLAB's Classification Learner toolbox were utilized in [14] to classify defects based on stator current and vibration signal data. Fine Gaussian SVM, Fine k-NN, Weighted k-NN, Bagged Trees, and Subspace k-NN were employed to achieve the almost perfect classification. However, large-scale industrial use and real-time applications were not investigated, which could impact deployment viability.

A CNN transfer learning method was proposed in [15] for centrifugal pump fault detection from Motor Current Signature Analysis (MCSA) data. Instead of synthetic laboratory test data, the investigation used actual operating pump data DQ/Concordia patterns. Data were gathered over two years with a distributed IoT system, and ResNet-34 was employed for faulty vs. non-faulty classification. The model registered an overall accuracy of 85.51 %, proving that CNNs can potentially detect real-world faults. Despite challenges posed by the limited access to real-world faulty datasets and data augmentation requirements,

challenges were identified. The research suggested edge-device deployment and real-time classification research in the future. Predictive maintenance models have been proposed [16,17]. In [16], vibration and temperature readings of four three-phase induction motors were used to create a predictive maintenance model. The dataset contained over 1.6 million data points and was preprocessed using noise removal and feature selection methods. K-NN, RF, DT, and Artificial Neural Network (ANN) classifiers were used; Random Forest produced the highest accuracy of 91 %. The research found that Random Forest was the most stable and generalizable model, with ANN showing good pattern recognition capabilities. Real-world deployment issues, more specifically in managing heterogeneous fault conditions, were, however, recognized. A predictive maintenance system based on ANN for seven electrical faults in induction motors was implemented in [17], utilizing real-time voltage and current signal measurements. A feed-forward backpropagation neural network trained with MATLAB's neural network toolbox was used. The ANN model could classify faults with almost 100 % accuracy and was implemented in an actual motor system through cloud-based monitoring. Although the study proved the superiority of ANN over traditional fault detection techniques, computational complexity and scalability problems in industrial applications were not investigated in detail.

Hybrid models integrating signal processing with machine learning have also been studied [18,19]. In [18], an induction motor fault diagnosis system was implemented using MATLAB Simulink. Four faults, namely open circuit, short circuit, overload, and faulty rotor bars, were simulated to generate one hundred fifty thousand data points with a healthy-to-faulty data split of 60–40 %. Some classifiers employed included Decision Trees, SVM, Logistic Regression, and Naive Bayes; among these, Decision Trees showed the highest accuracy (92 %). The study did not address real-time deployment issues but emphasized combining machine learning and signal processing for effective fault categorization. A genetic algorithm (GA) feature selection was introduced in [19] for bearing defect diagnosis of induction motors. K-NN, Random Forest, and Decision Tree classifiers were employed; the highest accuracy of 99.7 % was obtained by Random Forest. The GA-based feature selection utilized cut reduced significant computation complexity but retained the classification accuracy to an extremely high level. The work also identified advantages in utilizing the motor current signal in non-intrusive diagnosis but accepted its potential for expanded test runs at various operating conditions on different motors.

Software-in-the-loop simulation platforms have found applications in training machine learning algorithms in simulation data [20,21]. In [20], SiL-based simulations were used to identify incipient short-circuit faults (ISCF) by examining voltage signals due to axial leakage flux. Multiresolution analysis (MRA) by DWT and classifiers like SVM, k-NN, Optimum Path Forest (OPF), and Multi-Layer Perceptron (MLP) were tried with 99.23 % accuracy. The study demonstrated the potential of predictive maintenance systems based on machine learning for Industry 4.0 applications, but it also identified the drawback of reducing processing time for real-time monitoring. A new fault diagnosis model for three-phase induction motors using multiple sensor data sources was proposed in [21]. The research included a large-scale experimental test rig that simulated 11 health conditions with different severity levels. Data were captured via vibration sensors, non-contact proximity probes, and current sensors. Various machine learning algorithms, including SVM, Ensemble-Bagged Trees, Neural Networks, and Quadratic Discriminant Analysis, were attempted, and SVM yielded a mean overall accuracy greater than 90 %. Challenges of real-time deployment and future cross-validation on larger industrial-sized datasets were exposed in this work.

Lastly, applications in real-world scenarios with AI-based recommendations have been suggested for rotor bar fault diagnosis [22,23]. In [22], an AI-based model that could process up to 80 % white noise without compromising predictive performance was implemented. The research proved the strength of machine learning for industrial purposes

but indicated possible difficulties in modifying the model for various motor configurations. A fault detection system using the Auto-Regressive Neural Network with Exogenous Inputs (NARX) model was proposed in [23]. The model was adequate for predictive maintenance and good classification performance for different fault types. Still, the study found replicating the approach in various industrial environments challenging. These studies recognize the potential of machine learning and deep learning in induction motor fault diagnosis, noting that they can enhance the accuracy of defect detection and predictive maintenance. Scalability, real-time processing, and computational efficiency remain the main issues, necessitating further research, optimization techniques, and integration of advanced architectures for improved performance in industrial applications.

Deep learning-based techniques have been extensively investigated for diagnosing faults in electric motors and enhancing predictive maintenance [24,25]. In [24], Shannon, Renyi, and Tsallis entropy metrics were used to develop an entropy-based multi-class fault detection model to improve classification accuracy. The work included a Multi-Layer Perceptron Classifier (MLPC) that optimized hyperparameters using Optuna, reflecting enhanced computational performance and classification. Although the approach effectively classified different types of faults, the difficulty of adapting the model for real-time systems continues to persist. In [25], a DNN model was constructed to discriminate electrical and mechanical faults using current and vibration signals. Statistical features like standard deviation, kurtosis, and variance were utilized to extract features for enhancing the accuracy of classification. The findings indicated that the DNN model was superior to conventional machine learning methods, providing good accuracy in fault detection. The study noted the need for trading model complexity for computational feasibility when implemented in real-world industrial environments.

Subsequent research on enhancing augmented reality (AR) incorporation in fault diagnosis was investigated in [26]. An intelligent health monitoring system of the motor based on Long Short-Term Memory (LSTM) networks combined with AR visualization was proposed. Real-time understanding of motor condition was offered to enhance user experience and maintenance productivity. Although predictive maintenance was reported to show significant improvements in the research, drawbacks towards large-scale industrial applications as well as optimizing AR were realized. Comparative study of machine learning algorithms to diagnose traction motor faults was conducted in [27]. Various ML classifiers were evaluated on binary, multi-class, and multi-label fault classification. It was found that Random Forest and Neural Networks were highly competent in identifying faults, and SVM performed well with multi-class classification. Despite the success of these algorithms, the study highlighted the importance of choosing proper classification techniques for specific fault detection issues.

Artificial intelligence (AI) and machine learning have contributed significantly towards enhanced fault diagnosis and detection in electric machines. Fault detection generally involves handcrafted features and rule-based systems, which are limiting when it comes to scalability and real-time operation [28]. Multiple AI techniques for increasing the efficiency and accuracy of fault detection have been studied by research, including machine learning, deep learning, and hybrid techniques. In [28], Gawande et al. contrasted deep learning with conventional methods for the fault diagnosis of electrical machines. The study contrasted models such as deep neural networks (DNNs), support vector machines (SVMs), decision trees, and ensemble approaches on industrial datasets. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) outperformed conventional methods based on accuracy and reliability, according to their findings. However, the study pointed out challenges in terms of interpretability of deep learning models and computational capacity for real-time fault detection.

Ahmad et al. [29] have studied AI-based fault detection in electrical motors with an emphasis on machine learning algorithms like SVM, RF, and KNN. Based on the study, CNNs and LSTM networks were found to

deliver the best results in motor fault classification. Hybrid methods like CNN-LSTM models proved to be better than the others by performing feature extraction in spatial and temporal domains of motor signals. Nonetheless, the research underscored the importance of better data-gathering methods and feature extraction strategies in order to drive fault classification efficiency.

In [30], Shankar et al. proposed an intelligent fault diagnosis system using AI methods, such as ensemble methods Random Forest and Gradient Boosting. They used a combination of CNNs and RNNs to detect motor faults from sensor data. The research proved that deep learning models were capable of detecting subtle fault patterns efficiently, enhancing predictive maintenance approaches. Nevertheless, the authors pointed out that deep learning models need large labeled datasets for training, which is a challenge in real-world industrial applications.

Gonzalez-Jimenez et al. [31] proposed a machine learning fault detection system for induction machines via Software-in-the-Loop (SiL) simulations. The research was concerned with the diagnosis of high-resistance connections, open-phase faults, and reverse wiring connections of motor power terminals. The research proved the capability of SiL simulations in producing training data for AI models, minimizing the dependency on actual fault datasets. Although successful, the research recognized the drawback of simulated data, which cannot fully represent real-world operational variations.

G. V. Nagesh Kumar et al. [32] described a wind turbine system emulation model based on a vector-controlled induction motor drive (VCIMD) with a fuzzy logic controller. The method is designed to simulate real wind turbine operation without utilizing real wind, enabling proper simulation through electrical machines. Their approach uses both 1-DOF and 2-DOF control methods to emulate inertia and improves dynamic response through compensation of torque errors with a fuzzy controller. The paper employs MATLAB/Simulink as the simulation tool and confirms performance with experimental platforms comprising induction motors and doubly-fed induction generators (DFIG). Though the work here doesn't specifically handle fault diagnosis or machine learning application, it accentuates the criticality of a robust control method for variable-speed induction motors with insights into emulation stability, torque tracking, and disturbance rejection issues that are key to model-based fault detection frameworks such as in the

current work.

While the current study successfully employs machine learning classifiers for fault detection in three-phase induction motors using voltage, current, and speed features, it does not incorporate thermal parameters, such as temperature rise, which are critical for diagnosing incipient faults like rotor insulation damage or overload. Sudha et al. [33] demonstrated that incorporating temperature prediction significantly enhances the reliability of condition monitoring in Squirrel Cage Induction Motors. Their work used temperature as a continuous variable in ML models, enabling early-stage fault forecasting. In contrast, our classification approach remains static and does not leverage the temporal dynamics of thermal growth.

Furthermore, our system relies on simulation data, which, although controlled, lacks real-world validation against measured temperature rise under operational stress. This could limit model generalization. Additionally, thermal-electrical feature fusion—wherein both types of data are used in training is missing, which has been shown by Sudha et al. to improve model robustness. Lastly, while our work emphasizes classification, forecasting temperature evolution via regression or hybrid ML methods remains unexplored, presenting an avenue for future enhancement.

3. Simulation model

The process starts with data acquisition and preprocessing, where motor parameters including voltage, current, and speed are received from MATLAB/SIMULINK environment simulation. Mathematical model of 3-PIM is presented in Fig. 1. These data sets are annotated with fault conditions such as 'Healthy,' 'R-G,' 'RYB-G,' 'RYG,' 'Load Variation,' and various over-voltage conditions. Data was normalized to maintain uniformity and improve classifiers' performance. Next, feature selection was performed, wherein appropriate motor parameters associated with fault conditions were selected to decrease the dimensionality of the dataset and enhance the efficiency of the models.

The procedure starts with data acquisition and preprocessing, wherein motor parameters like voltage, current, and speed are collected from MATLAB/SIMULINK environment simulation. The mathematical model of 3-PIM is presented in Fig. 1. The induction motor considered in

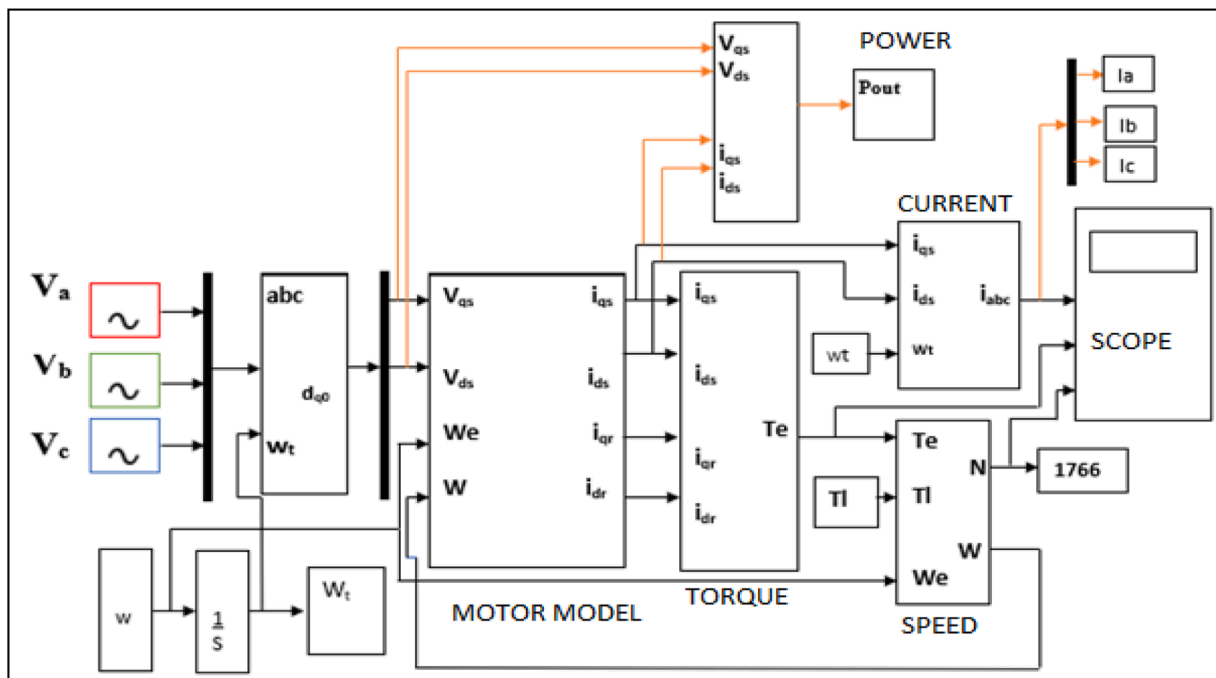


Fig. 1. Mathematical model of 3-PIM Similarly, the rotor voltage equations are.

this study has a rated power of 7.5 kW, operates at a frequency of 60 Hz, and is supplied with 220 V. The detailed motor parameters are provided in Table 1. These datasets are marked with some fault conditions, i.e., 'Healthy,' 'R-G,' 'RYB-G,' 'RYG,' 'Load Variation,' and various over-voltage conditions. The data was normalized to make it uniform and improve the performance of classifiers. Then, feature selection was performed where appropriate motor parameters that map to error states were chosen in order to reduce the dataset size and enhance the models' efficiency.

$$V_{ds} = R_s I_{ds} + \frac{d\lambda_{ds}}{dt} - \omega_s \lambda_{qs} \quad (1)$$

$$V_{qs} = R_s I_{qs} + \frac{d\lambda_{qs}}{dt} + \omega_s \lambda_{ds} \quad (2)$$

$$V_{dr} = R_r I_{dr} + \frac{d\lambda_{dr}}{dt} - (\omega_s - \omega_r) \lambda_{qr} \quad (3)$$

$$V_{qr} = R_r I_{qr} + \frac{d\lambda_{qr}}{dt} + (\omega_s - \omega_r) \lambda_{dr} \quad (4)$$

Where, V_{ds} , V_{qs} = Stator voltages in direct and quadrature axes

V_{dr} , V_{qr} = Rotor voltages in direct and quadrature axes

I_{ds} , I_{qs} , I_{dr} , I_{qr} = Rotor voltages in direct and quadrature axes

V_{dr} , V_{qr} = Stator and rotor currents

R_s , R_r = Stator and rotor resistances

λ_{ds} , λ_{qs} , λ_{dr} , λ_{qr} = Flux linkages

ω_s = Synchronous speed

ω_r = Rotor speed

The flux linkage equations define the interaction between stator and rotor magnetic fields:

$$\lambda_{ds} = L_s I_{ds} + L_m I_{dr} \quad (5)$$

$$\lambda_{qs} = L_s I_{qs} + L_m I_{qr} \quad (6)$$

$$\lambda_{dr} = L_s I_{dr} + L_m I_{ds} \quad (7)$$

$$\lambda_{qr} = L_r I_{qr} + L_m I_{qr} \quad (8)$$

Where,

L_s , L_r = Stator and rotor self-inductances

L_m = Mutual inductance

When faults arise, like resistance, inductance, or voltage variations, they directly impact these equations, resulting in anomalies in motor performance. Through the observation of deviations from normal behavior based on real-time sensor readings, faults can be identified effectively. The data set for nine types of faults is created by further research from this system.

4. Methodology

Using advanced machine learning techniques to categorize and identify numerous defects from key operating characteristics is the

foundation of the fault analysis methodology for three-phase induction motors. The procedure begins with data collection and preprocessing since the voltage, current, and speed sensor inputs are collected with various fault operating conditions. Subsequently, the data set is labeled with particular motor faults to achieve precise machine-learning algorithm training. Feature selection identifies the top-ranked parameters for fault classification, dimension reduction, and enhanced computation efficiency. Several machine learning algorithms, i.e., RF, SVM, KNN, GBM, and XGBoost + FIS, are used to train the models for prediction. All the models are hyperparameter-tuned to increase the accuracy and classifying potential. The merits and effectiveness of the models are then assessed and compared using various performance measures, such as accuracy, precision, recall, and F1-score. Ensemble learning is also investigated to enhance diagnostic reliability, and XGBoost is coupled with a FIS to encourage interpretability. Lastly, a detailed evaluation of all the models is performed to find the most accurate and efficient method for real-time fault detection and predictive maintenance of induction motors. The comparative analysis aids in determining the best machine-learning technique. The block diagram showing the fault recognition process of the Three-phase induction motor through a machine-learning algorithm is represented in Fig. 2, reflecting the systematic flow.

4.1. Dataset

The data employed here is a complete, 1600-sample dataset, and a sample is related to different operating conditions of a three-phase induction motor (3-PIM). The sample has nine critical features, which are vital in identifying the health of the motor and establishing possible faults. The features are major electrical and mechanical parameters such as:

- Direct Quadrature Current Component (i_{dq}) – The direct and quadrature axis current is needed to comprehend motor dynamics.
- Output Power (P_{out}) – The motor provides power under different load conditions.
- Motor-Produced Torque – The motor's torque level is inherently associated with load changes and mechanical efficiency.
- Rotational Speed (wt) – Indicates the speed of the motor shaft, varying with the applied load and fault conditions.
- Phase Currents (i_a , i_b , i_c) – The stator phase electrical currents A, B, and C are beneficial for imbalance and short-circuit fault detection.
- Input Power (P_{in}) – The sum of electrical power delivered to the motor is helpful for efficiency analysis and fault prediction.

The 10th column in the dataset is the target feature, the fault type label. It labels each sample as one out of eight conditions, with one being healthy and seven others being different types of faults. The fault types allow machine learning models to learn patterns for various fault conditions, which can improve fault classification accuracy.

4.2. Pre-processing

Preprocessing plays a significant role in preparing the 3-PIMs machine learning fault classification dataset. Preprocessing is initiated by cleaning data, which involves processing missing values through interpolation, mean imputation, and outliers caused by sensor failure or variation identified using Z-score analysis or interquartile range (IQR) methods and corrected or eliminated based on the circumstances. Standardization and normalization introduce equality between features and do not favor features with higher numerical values. While the Z-score standardization normalizes to 0 mean and 1 standard deviation to obtain equal distribution, the Min-max normalization scales features from 0 to 1 but maintains relative relationships. These operations facilitate machine learning algorithms to process features in the best possible way.

Table 1
Induction Motor Parameter.

Sr. No.	Parameter	Value	Symbol
1	Frequency	60 Hz	P_{rated}
2	Supply Voltage	220 V	F
3	Moment of Inertia	0.4 kg·m ²	V_m
4	Stator Resistance	0.288 Ω	J
5	Rotor Resistance	0.158 Ω	r_s
6	Stator Self-Inductance	42.5 mH	r_r
7	Rotor Self-Inductance	41.8 mH	L_s
8	Mutual Inductance between Stator and Rotor	41.2 mH	L_r
9	Rated Torque	40 N·m	L_m
10	Number of Poles	4	T_{rated}

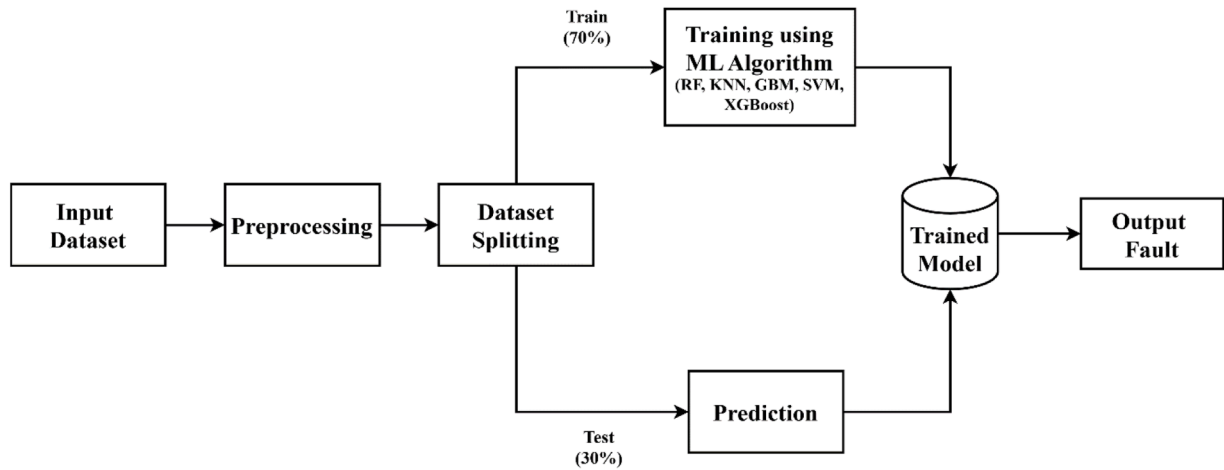


Fig. 2. Block diagram of the Three phase induction motor using a machine learning algorithm Each block of the block diagram is explained below.

Besides, advanced feature selection methods eliminate duplicated features, reduce dimensionality, and enhance computing efficiency. Such methods are Principal Component Analysis (PCA), Random Forest Feature Importance, and association analysis. These methods strengthen model performance by employing feature set optimization, improved performance, accuracy, and generalization on multiple datasets and machine learning tasks.

4.3. Dataset splitting

The data is separated into three sets to enable appropriate model training, validation, and testing: 70 % for training and 30 % for testing. The validation data is utilized for model selection and hyperparameter tuning to avoid overfitting or underfitting. In contrast, the training data reduces the chances that the machine learning models will find spurious patterns. Finally, the test set objectively measures model accuracy on unseen instances. For maintaining consistency, stratified sampling is applied, and each class (healthy state and seven fault classes) is assigned proportional representation within all subsets to prevent bias towards more frequent types of faults.

4.4. Training

To enable appropriate model training, validation, and testing, 90 % of the data is put aside for training, while 30 % is set aside for testing. In this phase, the training samples (70 % of the total) are made available to several machine-learning models to learn patterns and patterns between input features and fault classes. The algorithms employed here are RF, SVM, KNN, GBM, and XGBoost + FIS. All these algorithms are trained with hyperparameters optimized to peak performance for fault detection, which is discussed in detail later.

4.4.1. KNN

The K-Nearest Neighbors (KNN) algorithm is among the most commonly employed supervised learning methods for classification problems, including fault detection in three-phase induction motors. The algorithm is simple to apply and efficient [34]. The instance-based learning approach without parameters employs the K nearest data instances (neighbors) in the feature space to predict the class of a novel sample. KNN lacks an underlying training process; it makes predictions using new cases based on similarity metrics like Euclidean distance. Its accuracy relies on the value of K and the distance metric used, so optimizing these parameters for better accuracy in motor fault detection is essential. The classification uses a majority vote principle, where the new point is classified as the most frequent type of fault among the K nearest points [35,36]. In fault detection, KNN helps to analyze

real-time sensor readings, e.g., current, voltage, temperature, and vibration signals, and cross-correlate them with labeled past data to classify motor conditions into known states (healthy or fault states). Sample similarity is typically expressed with the Euclidean distance; however, there are alternatives in the form of Manhattan or Minkowski distances depending on the dataset.

The KNN approach determines the class label of the most recent data point by calculating the Euclidean distance between it and previously labeled data points. The Euclidean distance between a test sample $X = x_1, x_2, \dots, x_n$ and a training sample $X_i = x_{i1}, x_{i2}, \dots, x_{in}$ is computed as:

$$D(X, X_i) = \sqrt{\sum_{j=1}^n (x_j - x_{ij})^2} \quad (9)$$

Where,

$D(X, X_i)$ is the distance between the test sample X and the training sample X_i , x_i , x_{ij} are the feature values of the test and training samples, respectively, n represent the total number of features (e.g., current, voltage, torque, rotational speed, etc.)

The K nearest neighbors are chosen based on the shortest distance values after the distances for each training sample have been determined. A majority vote chooses Class $C(X)$. The majority vote approach assigns the class label based on the most prevalent class among the K-selected neighbors:

$$C(K) = \arg \max_{c_k} \sum_{i=1}^K I(c_i = c_k) \quad (10)$$

4.4.2. SVM

Support Vector Machine (SVM) is a popular supervised learning classifier for many classification problems, such as fault diagnosis in three-phase induction motors. Its fundamental idea is to find the best hyperplane in an n-dimensional feature space that maximally separates two classes of faults. SVM tries to maximize the margin between two classes to minimize misclassification and enhance generalization. It makes use of sensor readings like voltage, current, vibration, and temperature to measure motor conditions and decide if they are normal or defective. SVM linearly separates data that is not linearly separable through the use of kernel functions. Some of the popular kernel functions are radial basis function (RBF), polynomial, linear, and sigmoid kernels [37]. SVM tries to identify the best hyperplane with the highest margin between two classes. The hyperplane equation in an n-dimensional feature space is represented by:

$$\omega_T x + b = 0 \quad (11)$$

Where ω is the weight vector that determines the orientation of the hyperplane, x is the input feature vector (sensor data such as current, voltage, torque, etc.), and b is the bias term.

For classification, the decision function is given by:

$$f(x) = \text{sign}(\omega_T x + b) \quad (12)$$

If f is more significant than 0, the sample belongs to one class (e.g., healthy motor), whereas if f of x is less than 0,

$f(x) < 0$, it belongs to another class (e.g., fault detected).

To maximize the margin M , SVM solves the following optimization problem:

$$\min_{\omega, b} \frac{1}{2} \|\omega\|^2 \quad (13)$$

Subject to the constraints:

SVM utilizes the kernel functions to map input features into higher-dimensional space to separate non-linearly separable data. Frequently used Radial Basis Function (RBF) kernel is provided by:

$$K(x, x') = e^{-\gamma \|x - x'\|^2} \quad (14)$$

Gamma is a hyperparameter that controls the influence of a single training sample.

4.4.3. Gradient boosting machines (GBM)

Gradient Boosting Machines (GBM), an ensemble learning technique, builds numerous decision trees sequentially, one upon another, each attempting to rectify the mistakes of the preceding ones. Being highly accurate and able to handle complex data sets, GBM is mostly used in three-phase induction motor fault detection [38]. The key concept of GBM is to use weak learner forecasting power (typically decision trees) to construct a strong learner. GBM in fault diagnosis processes sensor readings (voltage, current, speed, temperature) and progressively enhances classification by paying more attention to mislabeled instances. By minimizing a loss function with gradient descent, GBM ensures that each tree reduces the residual errors from the previous model and leads to the best classification model.

GBM minimizes a pre-defined loss function by iteratively improving weak learners. Given a training dataset $D = \{x_i, y_i\}$, where x_i represents feature vectors (sensor data) and y_i represents fault labels, GBM starts with an initial weak model:

$$F_0(x) = \arg \min_{\gamma} \sum_{i=1}^N L(y_i, \gamma) \quad (15)$$

where $L(y_i, \gamma)$ is the loss function (e.g., log loss for classification). At each iteration m , a new decision tree $h_m(x)$ is trained to approximate the negative gradient of the loss function:

$$h_m(x) = -\nabla_{F_{m-1}} L(y_i, F_{m-1}(x_i)) \quad (16)$$

The new model is updated as follows:

$$F_m(x) = F_{m-1}(x) + \lambda h_m(x) \quad (17)$$

Where λ is the learning rate that controls the contribution of each tree to the final model.

4.4.4. Random forest

Multiple decision trees are combined in Random Forest (RF), a strong ensemble learning technique, to increase classification accuracy and robustness. Since RF can handle complex, high-dimensional data and minimize overfitting, it is frequently utilized for fault diagnostics of three-phase induction motors. RF uses random data samples to train several decision trees, each producing a separate prediction about the fault status, which is then determined by a majority voting process [39].

RF processes sensor signals like voltage, current, torque, and speed for fault diagnosis to label motor conditions as healthy or faulty. Being an ensemble, RF can effectively deal with noise and missing values and

give robust and reliable predictions.

Each of the several Random Forest decision trees is built using a random training data selection. Given a training dataset $D = \{x_i, y_i\}_{i=1}^N$, where x_i represents sensor feature vectors and y_i represents fault labels, RF builds T decision trees. To ensure diversity within the training data, bootstrap sampling trains every decision tree within the ensemble on a randomly selected subset of training examples with replacement. Random feature selection at every tree node is also utilized to enhance model generalization and reduce trees' correlation. A random subset of features is used to decide on the best split. Finally, the ensemble employs majority voting for classification problems, which enhances the accuracy and robustness offered by aggregating the predictions of all trees and outputting the class with the highest votes as the final result.

$$C_{final}(x) = \arg \max_{c_k} \sum_{t=1}^T I(C_t(x) = c_k) \quad (18)$$

Where, $C_t(x)$ is the class prediction from tree t , c_k represents different fault types, $I(C_t(x) = c_k)$ is an indicator function that returns 1 if the tree t classifies the input as class c_k , otherwise 0.

4.4.3. Extreme gradient boosting with feature interaction (XGBoost + FIS)

An improved gradient boosting technique called Extreme Gradient Boosting (XGBoost) learns by gradually adding decision trees to a model to increase performance by lowering errors from previous iterations. XGBoost is efficient, scalable, and popularly used to solve classification problems such as fault detection in three-phase induction motors. However, while XGBoost offers high accuracy, its black-box nature interprets challenging [40]. Feature Interaction with Fuzzy Inference System (FIS) is integrated with XGBoost to enhance interpretability. FIS applies rule-based reasoning to extract insights from the predictions, improving fault classification transparency. This hybrid approach leverages XGBoost's predictive power and FIS's human-readable rules, providing an explainable and accurate fault detection system. The flowchart of the proposed XGBoost-FIS system is shown in Fig. 3.

Given a dataset $D = \{x_i, y_i\}_{i=1}^N$, where x_i are sensor readings and y_i are fault labels, XGBoost minimizes a loss function using gradient boosting:

$$L(F_m) = \sum_{i=1}^N L(y_i, F_{m-1}(x_i) + h_m(x_i)) + \Omega(h_m) \quad (19)$$

Where, F_{m-1} is the ensemble model at iteration m , $h_m(x_i)$ is the new tree model, $\Omega(h_m)$ is a regularization term to control model complexity. The model improves by updating trees using the gradient of the loss function.

$$h_m(x) = -\nabla_{F_{m-1}} L(y_i, F_{m-1}(x_i)) \quad (20)$$

Once XGBoost identifies the importance of features, FIS applies rule-based reasoning to enhance interpretability. A fuzzy rule in FIS is defined as:

$$\text{IF}(\text{Feature}_1 \text{ is High}) \text{ AND } (\text{Feature}_2 \text{ is LOW}) \text{ THEN } (\text{FaultType} = \text{ClassA}) \quad (21)$$

Fuzzy membership functions define the degree of belongingness:

$$\mu_{Low(x)} = \frac{1}{1 + e^{-\alpha(x-c)}} \quad (22)$$

where α and c are parameters control the smoothness and center of the membership function, the final decision rule combines XGBoost predictions with FIS reasoning, ensuring a more interpretable fault detection process.

By providing high accuracy through XGBoost and interpretability through FIS, XGBoost + FIS delivers a combined solution for fault detection in three-phase induction motors. The solution enhances predictive maintenance by providing interpretable fault classification,

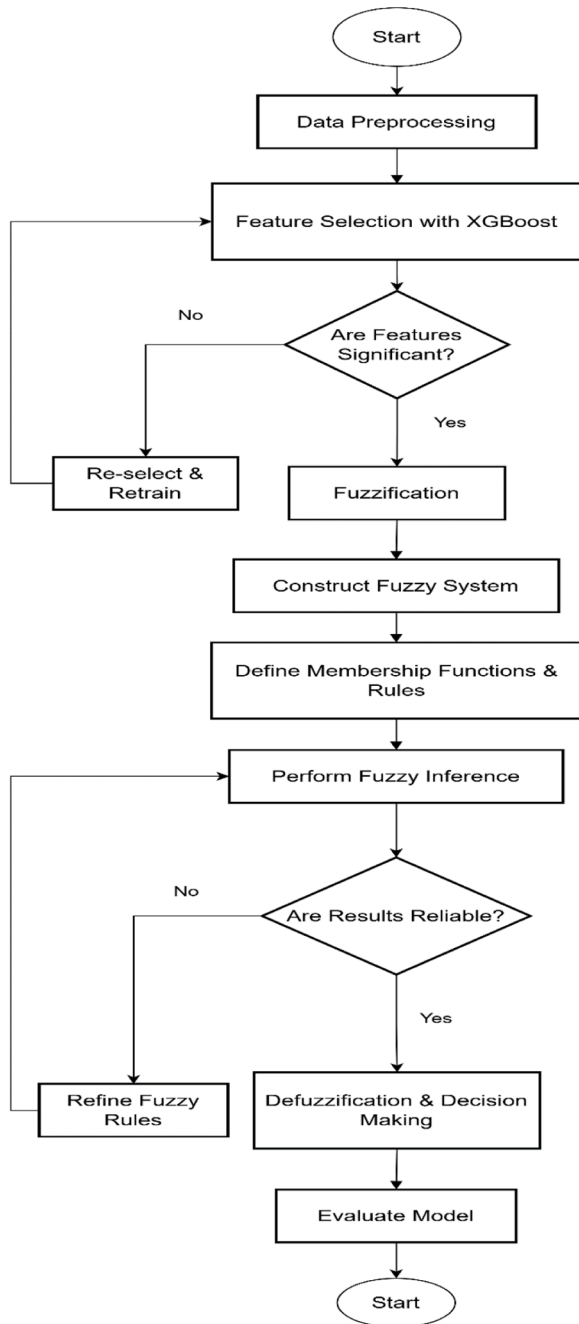


Fig. 3. Flowchart of proposed XGBoost-FIS system.

enabling industries to detect faults early while experiencing the reasoning behind the predictions. Using gradient boosting and fuzzy logic makes the solution very effective in complex industrial fault diagnosis applications.

Performance evaluation

Precision, accuracy, recall, and F1-score are used to decide the performance of machine learning algorithms in detecting faults in three-phase induction motors. Recall assesses the capacity to detect errors, accuracy assesses overall accuracy, precision assesses the proportion of accurately predicted faults, and F1-score is the harmonic mean of recall and precision. Other measures, such as confusion matrices, ROC-AUC curves, and execution time, offer more insights. It is also analyzed for performance with feature selection and preprocessing through model

comparison between SVM, RF, ANN, and CNN to find the best fault detection method.

The ratio of accurately detected defective motors to all anticipated defective cases is known as precision.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (23)$$

The higher the precision, the fewer false alarms are in fault detection.

- Recall: Recall is the ratio of actual faulty motors correctly classified

$$\text{Recall} = \frac{TP}{TP + FN} \quad (24)$$

The higher the recall, the fewer the missed fault cases.

- F1-Score: The F1-Score, the harmonic mean of precision and recall, provides a reasonable metric when the costs of false positives and negatives vary.

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (25)$$

- Accuracy: Accuracy is a metric that quantifies how accurately the model predicts outcomes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (26)$$

TP (True Positive): Properly predicted defective motors,
 TN (True Negative): Properly predicted healthy motors,
 FP (False Positive): Mispredicted defective motors,
 FN (False Negative): Mispredicted healthy motors.

Performance testing plays an important role when choosing the optimal model to detect three-phase induction motor faults. The most efficient algorithm for real-time predictive maintenance can be found using basic metrics like accuracy, precision, recall, and F1-score. Among the tested models, XGBoost + FIS presents a trade-off between interpretability and high accuracy and thus can be considered appropriate for industrial application.

5. Results & discussion

The performance assessment of the machine learning methods used for three-phase induction motor failure detection is provided in the Results and Discussion section.

5.1. Performance of machine learning algorithms for classification of fault in three-phase induction motor

The performances of algorithmic models like RF, KNN, GBM, SVM, and XGBoost + FIS are compared with key performance metrics like accuracy, precision, recall, F1-score, and ROC-AUC analysis. The obtained results give the performance of all models in identifying various motor faults and faulty and normal conditions of the motor. The comparison also decides the advantages and disadvantages of all models in terms of computational cost, noise resistance, and interpretability. Comparative analysis identifies the most suitable algorithm for best results to obtain an optimal solution for real-time predictive maintenance and industrial fault diagnosis. The result of each ML algorithm for fault recognition in the three-phase motor is provided below.

5.1.1. Result of K-Nearest neighbors for fault analysis in 3-Phase induction motors

KNN is one of the simplest and best machine-learning techniques for categorizing three-phase induction motor failures. The system categorizes defects based on similarities by comparing motor operating parameters like voltage, current, and speed to previously classified data. It is susceptible to distance measurement and feature scaling because it uses the majority class of the closest neighbors to classify flaws. Table 2 and Fig. 4 show the outcome of the KNN algorithm for 3-PIM fault detection.

Table 2 shows the K-Nearest Neighbors (KNN) algorithm fault classification performance for three-phase induction motor faults according to each class's precision, recall, F1-score, and support. The classification performance is average and has an overall accuracy of 61 %. The best performance among all the classes is achieved in Class 3 with precision = 0.87, recall = 1.00, and F1-score = 0.93, indicating that this fault is classified very accurately. Class 8 also has good performance with precision = 0.71 and recall = 0.89, giving an F1-score of 0.79, indicating high classification power for this type of fault. Classes 1, 2, and 5, on the other hand, have lower performance with F1-scores of 0.44, 0.42, and 0.46, respectively, suggesting that these fault classes are harder to distinguish, perhaps because they share overlapping feature distributions. Classes 4, 6, and 7 also exhibit moderate classification power with F1 scores between 0.58 and 0.61, demonstrating that these faults are subject to some misclassification. The uneven precision and recall values over various classes indicate the weaknesses of KNN in dealing with overlapping fault classes, which can be enhanced using optimized distance measures, feature subset selection, or ensemble methods. The performance metrics of KNN are graphically represented, as shown in Fig. 4, clearly indicating the classification efficiency and the disparity in detection accuracy over various faults.

5.1.2. Result of SVM algorithm for fault analysis in 3-Phase induction motors

The SVM algorithm classified errors in three-phase induction motors according to voltage, current, and speed. Accuracy, precision, recall, and F1-score were the performance metrics to test. The SVM model's performance for 3-PIM fault diagnosis can be seen in Table 3 and Fig. 5.

The overall accuracy of the SVM model was 68.12 %, reflecting a moderate classification ability. Class 2 and Class 3 were the best fault classes, with F1 scores of 0.93 and 0.80, respectively. The model showed high precision (1.00) for Classes 2, 4, and 5. Class 8 also showed a high recall of 0.98, reflecting good detection of this fault. Nonetheless, the SVM model encountered difficulties classifying some fault types, especially for Class 1 (Healthy motor) and Class 6 with F1-scores of 0.29 and 0.38, respectively.

It indicates that the model had difficulty differentiating between faulty and non-faulty states, possibly because fault categories share similar features. In addition, Class 4 and Class 5, with their ideal precision (1.00), had poorer recall values (0.54 and 0.62, respectively), suggesting that some real fault cases were misclassified or missed.

The graphical plot in Fig. 5 emphasizes the differences in classification performance between fault classes, which disclose an imbalance

Table 2
Performance for K-Nearest Neighbors.

Classes	Precision	Recall	f1-score	support
1	0.40	0.50	0.44	40
2	0.50	0.36	0.42	47
3	0.87	1.00	0.93	40
4	0.56	0.62	0.59	39
5	0.59	0.38	0.46	42
6	0.69	0.50	0.58	40
7	0.54	0.70	0.61	27
8	0.71	0.89	0.79	45
Accuracy				0.61

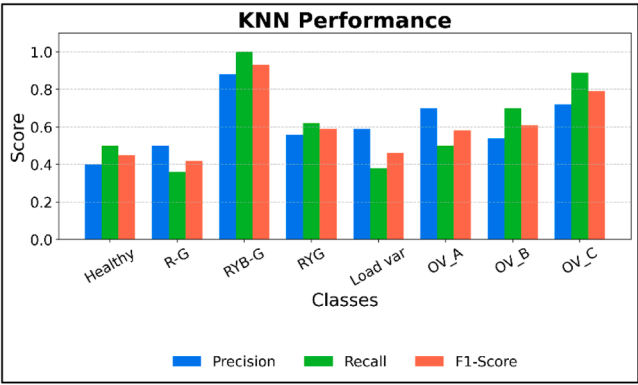


Fig. 4. Performance matrices for K-Nearest.

Table 3
Performance for Support Vector Machine.

Classes	Precision	Recall	f1-score	support
1	0.41	0.23	0.29	40
2	1.00	0.87	0.93	47
3	0.67	1.00	0.80	40
4	1.00	0.54	0.70	39
5	1.00	0.62	0.76	42
6	0.45	0.33	0.38	40
7	0.48	0.89	0.62	27
8	0.62	0.98	0.76	45
Accuracy				0.6812

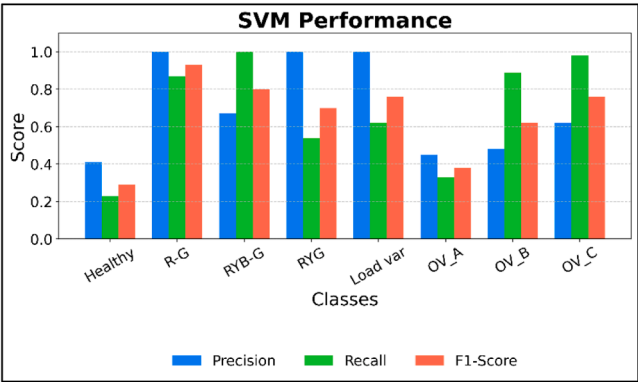


Fig. 5. Performance of SVM algorithm for three-phase induction motor fault recognition.

of precision and recall for various fault types. The incorrect classification of the standard motor state and some faults indicate more model optimization requirements. Technics like feature engineering, hyper-parameter adjustment, and kernel tuning (for example, using an RBF or polynomial kernel) may be applied to enhance the SVM model's accuracy. In addition, integrating fuzzy inference systems (FIS) into SVM can improve the interpretability and robustness of the model when addressing real-world fault detection issues. While the SVM algorithm performed exceptionally well in detecting motor faults, its lower accuracy relative to other models, such as Gradient Boosting (GBM) and Random Forest (RF), indicates improvement.

5.1.3. Result of GB algorithm for fault analysis in 3-Phase induction motors

TheGB algorithm classified 3-phase induction motor faults based on voltage, current, and speed. Its accuracy, measured through accuracy, recall, and F1-score, quantified fault detection efficiency. Results are reported in Table 4 and Fig. 6.

Table 4
Performance for Gradient Boosting Machine.

Classes	Precision	Recall	f1-score	support
1	0.89	0.85	0.87	40
2	0.98	0.96	0.97	47
3	0.98	1.00	0.99	40
4	1.00	0.95	0.97	39
5	0.93	0.90	0.92	42
6	0.84	0.78	0.81	40
7	0.76	0.96	0.85	27
8	0.93	0.96	0.95	45
Accuracy				0.92

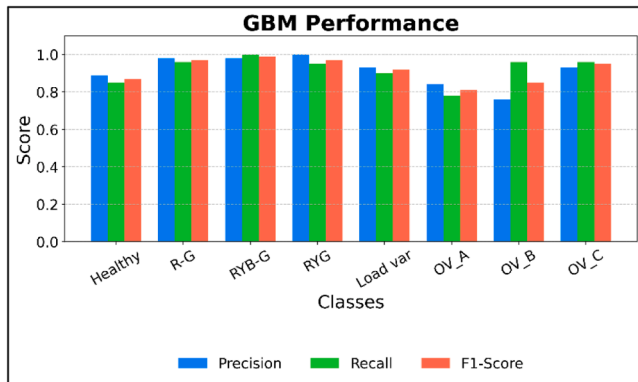


Fig. 6. Performance of GB algorithm for three-phase induction motor fault recognition.

The GB model attained a total accuracy of 92 %, proving its ability to classify various motor faults accurately. Class 3 and Class 4 of the fault classes performed the best, with F1 scores of 0.99 and 0.97, respectively. Moreover, Class 2 and Class 8 also did a great job with F1 scores of 0.97 and 0.95, respectively, showing that the model effectively classified most of these fault states with very few misclassifications. The close-to-1.00 precision values for several classes imply that the model made only a few wrong optimistic predictions so that the correct faults were well identified. Nevertheless, Class 6 and Class 7 showed relatively poorer classification performance with F1 scores of 0.81 and 0.85, respectively. Class 6's recall value of 0.78 was the lowest for all categories, which indicates that certain defects in this class went undetected by the model. Class 7 also exhibited a lower precision of 0.76, which implies some misclassifications where faults of other classes were wrongly predicted as Class 7.

The graphical plot in Fig. 6 emphasizes the GB model's good classification accuracy in most fault categories, enhancing its credibility for motor fault diagnosis. In contrast to other models like SVM (68.12 % accuracy), the GB algorithm improved accuracy and fault classification strength. Because of its boosting potential, it sequentially enhances classification by focusing on difficult-to-classify samples. To improve the model's performance, more research may be done on hyperparameter fine-tuning, including learning rate, number of estimators, and maximum depth of trees. Additionally, using GB with fuzzy inference systems (FIS) or an ensemble learning method can improve interpretability and classification accuracy. The GB algorithm was very efficient in diagnosing 3-phase induction motor faults and had a high accuracy rate of 92 %. Its usability in processing multiple and complicated fault cases is advantageous as a substitute for industrial predictive maintenance equipment. Its performance levels can be improved for difficult classes like Class 6 and Class 7 faults through improved feature selection strategies and optimization processes.

5.1.4. Result of RF algorithm for fault analysis in 3-Phase induction motors

An RF algorithm was applied to identify the fault of 3-phase

induction motors through important operating factors like voltage, current, and speed. Accuracy, recall, F1-score, and global accuracy were all used to measure the model's performance as a metric describing how effectively the model classified the faults. The performance of the RF algorithm for fault identification of 3-PIM is tabulated in Table 5 and Fig. 7.

The RF model recorded a total accuracy of 93.12 %, which reflects its stability and consistency in identifying motor faults under different categories. Class 3 and Class 2 performed the best among all the faults, recording an F1-score of 0.99 and 0.98, respectively, reflecting outstanding classification accuracy. Class 4 and Class 8 also showed strong performance at an F1-score of 0.97, confirming the ability of the model to distinguish these faults with minimal misclassification.

The high precision values (>0.95) in multiple classes reflect that the model made extremely low false optimistic predictions with high accuracy for fault identification. However, Class 1 (Healthy motor) and Class 6 demonstrated slightly lower classification performance, with F1 scores of 0.86 and 0.84, respectively. The recall for Class 1 was 0.78, which indicates that some healthy motors were misclassified as faulty. Similarly, Class 7 had a lower precision (0.77), but its recall was perfect (1.00), meaning that while all actual Class 7 faults were detected, some instances were misclassified into Class 7.

The graphical illustration in Fig. 7 also indicates the RF model's high classification precision, corroborating its appropriateness in motor fault detection. In contrast to other models like SVM (68.12 % accuracy) and Gradient Boosting (92 % accuracy), Random Forest ranked better in accuracy as well as robustness of classification, quite possibly because its ensemble learning feature lessens variance and enhances generalization. To further improve the model's performance, fine-tuning hyperparameters like the number of estimators, maximum depth of the tree, and minimum samples per split can be investigated. Furthermore, combining RF with feature selection methods can assist in minimizing feature redundancy, improving model efficiency further. The RF algorithm was very effective in fault detection of 3-phase induction motors with a noteworthy accuracy of 93.12 %. Its ability to perform multi-class classification with high precision and recall makes it an algorithm with promising potential for real-time predictive maintenance in industries.

5.1.5. Result of XGBoost+FIS hybrid model algorithm for fault analysis in 3-Phase induction motors

The XGBoost + FIS hybrid model was used for 3-phase induction motor fault detection, taking advantage of Extreme Gradient Boosting (XGBoost) in high predictive precision and Fuzzy Inference System (FIS) in interpretability. Precision, recall, F1-score, and accuracy were employed for the model to be assessed, and the outcome was that it could classify faults perfectly. GB algorithm performance for fault detection of 3-PIM is shown in Table 6 and Fig. 8.

The XGBoost + FIS model had an overall accuracy of 95 % and was the highest-performing model among all the algorithms tested. Of the various fault classes, Class 3 and Class 4 had perfect classification performance, both of which obtained an F1-score of 1.00, implying that the model did not commit any mistakes in predicting these fault types. The results of Class 2 and Class 7 were equally good, with F1 scores of 0.98

Table 5
Performance for Random Forest.

Classes	Precision	Recall	f1-score	support
1	0.97	0.78	0.86	40
2	0.98	0.98	0.98	47
3	0.98	1.00	0.99	40
4	1.00	0.95	0.97	39
5	0.93	0.95	0.94	42
6	0.85	0.82	0.84	40
7	0.77	1.00	0.87	27
8	0.96	0.98	0.97	45
Accuracy				0.93125

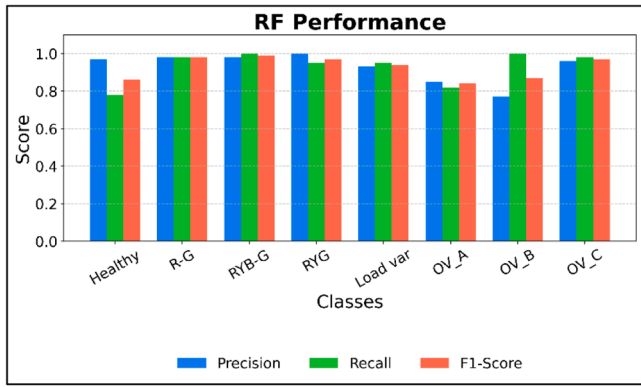


Fig. 7. Performance of RF algorithm for three-phase induction motor fault recognition.

Table 6
Performance for XGBoost + FIS hybrid model.

Classes	Precision	Recall	f1-score	Support
1	0.90	0.90	0.90sss	41
2	1.00	0.96	0.98	48
3	1.00	1.00	1.00	47
4	1.00	1.00	1.00	35
5	1.00	0.77	0.87	35
6	0.91	0.98	0.95	44
7	0.97	1.00	0.99	36
8	0.85	1.00	0.92	34
Accuracy				0.95

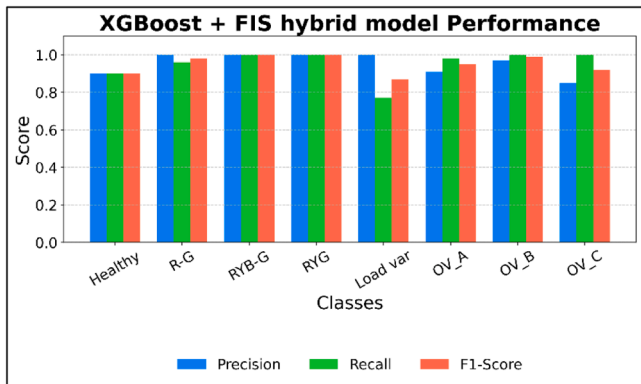


Fig. 8. Performance of XGBoost + FIS hybrid model algorithm for three-phase induction motor fault recognition.

and 0.99, respectively. A high precision of 1.00 across many classes indicates that the model kept false positives to an absolute minimum while reliably identifying faults. Class 5 did experience a low recall of 0.77, which translates into not identifying all instances of this type of fault. Class 8 also showed an accuracy of lower precision, 0.85, which means it misclassified some samples from another class as Class 8. Apart from these slight shortcomings, the model performed better than other methods like Random Forest (accuracy of 93.12 %) and Gradient Boosting (accuracy of 92 %), demonstrating better fault classification capability.

The graphical illustration in Fig. 8 emphasizes the model's high F1-score, recall, and precision across various fault classes, which supports its strength for fault detection in industrial motors. The combination of XGBoost with FIS balanced interpretability and predictive accuracy, making it an extremely efficient solution for real-time motor health monitoring. To further tune the model to improve performance, possible

enhancements can be fine-tuning the rule set of FIS to enhance decision-making performance and tuning parameters like learning rate and tree depth to further polish classification accuracy. Furthermore, increasing the dataset through additional fault variation could enhance the generalization ability for broader industry applications. The XGBoost + FIS hybrid model registered the highest classification accuracy of 95 % among all models tried and was, therefore, the best method for fault detection in the 3-phase induction motor.

5.2. Comparative analysis of performance of ML algorithms for classification of fault in 3-PIM

Three-phase induction motors are crucial in industries but tend to develop stator winding faults, rotor faults, and voltage imbalances, causing inefficiency and downtime. Conventional methods of fault detection are generally limited; ML is an effective tool for real-time fault classification. This subsection compares ML algorithms RF, KNN, SVM, GBM, and XGBoost+FIS using accuracy, precision, recall, and F1-score. A comparison of machine learning algorithms for the fault detection of 3-PIM is illustrated in Fig. 9.

There are noticeable differences in performance at precision, recall, F1-score levels, and accuracy in comparing defect diagnosis machine learning models for three-phase induction motors. Having the lowest rate of false negatives and the highest precision, recall, and F1-score, indicating its better capability to identify errors correctly, XGBoost + FIS consistently performs better than all the other models. Random Forest (RF) and Gradient Boosting Machine (GBM) also perform outstanding classification, with RF being very accurate and GBM being very recall, i.e., their power to reduce misclassification errors. SVM and KNN perform low recall and F1 values, showing that they do not handle complex multi-fault situations, which results in more undetected faults. Accuracy-wise, SVM and KNN are less precise, and GBM (92 %), RF (93 %), and XGBoost + FIS are the most accurate at 95 %. Overall analysis verifies that ensemble-based models, especially XGBoost + FIS, give the most accurate and interpretable fault classification, making them most appropriate for real-time predictive maintenance in industrial motor systems.

6. Conclusion

A comparison of three-phase induction motor failure diagnosis using machine learning models verifies the growing relevance of predictive maintenance driven by data in the industrial environment. The analysis compares specific ML models such as RF, KNN, SVM, GBM, and XGBoost + FIS based on significant performance measures such as precision, recall, F1-score, and accuracy. Among them, XGBoost + FIS has the maximum accuracy (95 %) and the best balance of recall and precision. The union of a Fuzzy Inference System (FIS) and XGBoost yields better classification and interpretability and is the best solution for fault classification. Results indicate that ensemble models (RF, GBM, and XGBoost) are superior to standard classifiers (SVM and KNN) because they handle multi-class fault scenarios efficiently, reduce misclassification error, and allow for varying motor conditions. SVM and KNN perform poorly in terms of precision and recall, which increases misclassification and, hence, is less desirable for industrial deployment in real-time. Also, the RF model has high fault detection accuracy (93 %) and excellent generalization power, which is a potential alternative. GBM, too, works well (92 %) but could need to be optimized further to address computational complexity and overfitting issues. This research confirms the efficacy of ML-based predictive maintenance solutions compared to conventional threshold-based diagnostic techniques. Automation of fault detection reduced human intervention and improved real-time decision-making capabilities, making machine learning an invaluable tool for industrial motor health monitoring. The results verify that sophisticated ML algorithms considerably enhance fault classification accuracy, minimize downtime, and maximize

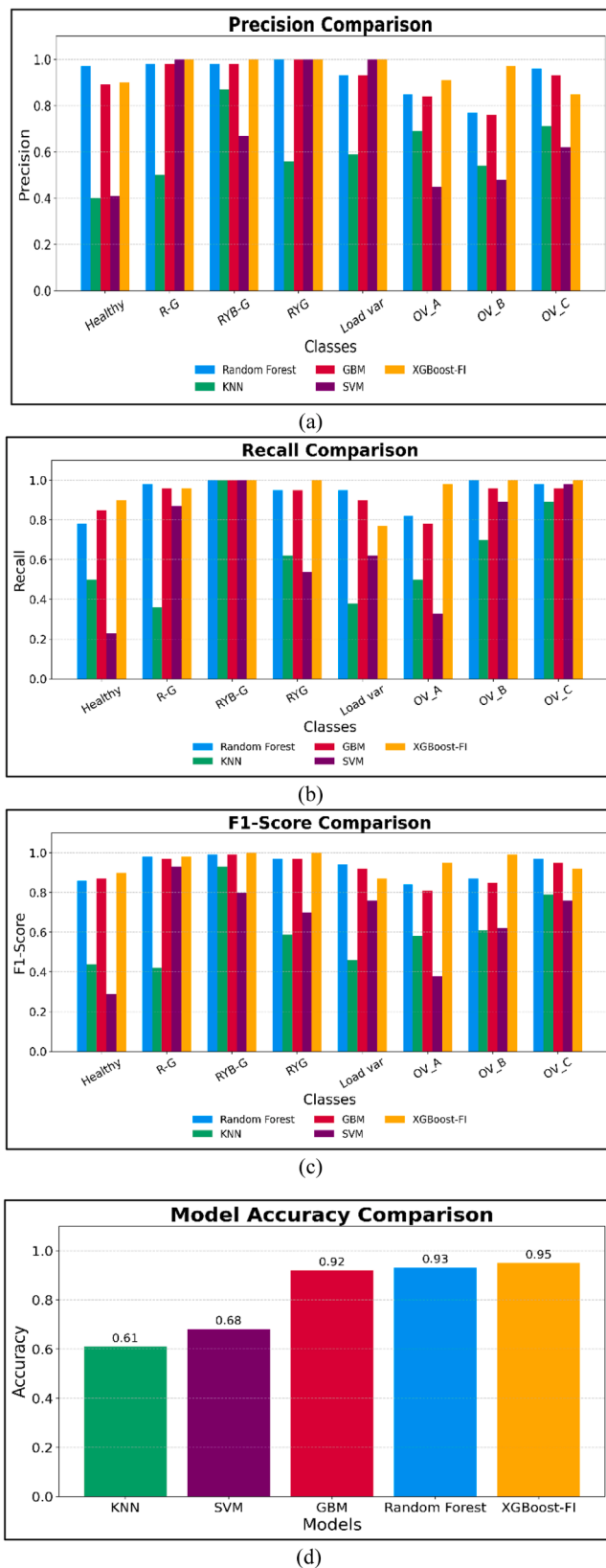


Fig. 9. Comparative analysis of ML algorithms for fault recognition in three-phase induction motors in terms of (a) Precision, (b) Recall, (c) F1 Score, (d) Accuracy.

maintenance strategies, ultimately resulting in cost savings and increased operational reliability.

Although the outcome proves the dominance of XGBoost + FIS, RF, and GBM, there is still room for improvement to make them more applicable in real-time, scalable, and computationally efficient. One of the prosperous fields for work in the future is the use of deep learning structures such as CNNs and Recurrent Neural Networks (RNNs), which can automatically learn raw motor data features and reduce the labor of feature engineering. These models could improve fault categorization by capturing more complex sensor data patterns. Creating hybrid models for real-time remote industrial motor monitoring that combine machine learning and Internet of Things (IoT) frameworks would be an additional avenue. Edge computing and cloud-based analytics can examine motor condition data, allowing for proactive maintenance scheduling and real-time defect identification. In addition, explainable AI (XAI) techniques can be explored to enhance model interpretability further and enable industry practitioners to gain better insights into fault classifications. Augmenting the dataset with actual industrial motor failure scenarios and implementing adaptive learning algorithms will also enhance model robustness. Utilizing transfer learning to transfer pre-trained models for specific motor fault detection applications could shorten training time without sacrificing accuracy. In addition, adding reinforcement learning for dynamic decision-making in predictive maintenance can fine-tune fault detection strategies in complicated industrial settings.

Last, using energy-efficient AI models is vital for deployment since huge ML models demand extensive computational resources. Future studies should aim to develop lightweight ML models that are deployable on embedded systems and industrial controllers to ensure real-time and low-latency fault diagnosis with minimal power usage. Through resolving these challenges, ML-based fault detection systems are made scalable, cost-competitive, and more accessible to industrial automation, thereby improving motor reliability and reducing unplanned downtimes.

CRedit authorship contribution statement

Puja Pohakar: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ravi Gandhi:** Writing – review & editing, Supervision, Methodology, Investigation. **Surender Hans:** Writing – review & editing, Supervision, Software. **Gulshan Sharma:** Writing – review & editing, Visualization, Validation, Resources. **Pitshou N. Bokoro:** Writing – review & editing, Supervision, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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